BOARD QUESTION PAPER: MARCH 2015 CHEMISTRY

Time: 3 Hours Total Marks: 70

Note:

i. **All questions are compulsory.**
ii. **Answers to the two sections are to be written in the same answer book.**
iii. **Figures to the right hand side indicate full marks.**
iv. **Write balanced chemical equations and draw neat and labelled diagrams, wherever necessary.**
v. **Use of logarithmic table is allowed.**
vi. **Answer to every question must be started on a new page.**

SECTION – I

Q.1. Select and write the most appropriate answer from the given alternatives for each sub-question: [7]

(i) p-type semi-conductors are made by mixing silicon with impurities of _______.

(a) Germanium (b) boron (c) arsenic (d) antimony

(ii) Amongst the following, identify the criterion for a process to be at equilibrium.

(a) $\Delta G < 0$ (b) $\Delta G > 0$ (c) $\Delta S_{total} = 0$ (d) $\Delta S < 0$

(iii) Colligative property depends only on _______ in a solution.

(a) Number of solute particles (b) number of solvent particles (c) nature of solute particles

(d) Nature of solvent particles

(iv) The charge of how many coulombs is required to deposit 1.0 g of sodium metal (molar mass 23.0 g mol^{-1}) from sodium ions?

(a) 2098 C (b) 96500 C (c) 193000 C (d) 4196 C

(v) What is the chemical composition of malachite?

(a) $CuO.CuCO_3$ (b) $Cu(OH)_2.CuCO_3$ (c) $CuO.Cu(OH)_2$ (d) $Cu_2O.Cu(OH)_2$

(vi) The element that does NOT exhibit allotropy is _______.

(a) As (b) Sb (c) Bi (d) N

(vii) The integrated rate equation for first order reaction A → products is _______.

(a) $K = 2.03t \log_{10}\frac{[A]_0}{[A]_t}$ (b) $K = \frac{1}{t}\ln\frac{[A]_t}{[A]_0}$

(c) $K = \frac{2.303}{t}\ln\frac{[A]_t}{[A]_0}$ (d) $K = \frac{1}{t}\ln\frac{[A]_t}{[A]_0}$

Solution 1:

(i) (b) Boron.

(ii) (c) $\Delta S_{total} = 0$

(iii) (a) Number of solute particles

(iv) (d) 4196

(v) (b) $Cu.(OH)_2CuCO_3$

(vi) (c) Bi

(vii) (c) $K = \frac{2.303}{t}\log_{10}\frac{[A]_t}{[A]_o}$

Q2. Answer any SIX of the following: **[12]**

(i) Define the following terms:

a. Enthalpy of fusion

b.Enthalpy of atomization

(ii) Derive van’t Hoff general solution equation.

(iii) Explain impurity defect in stainless steel with diagram.

(iv) Derive the relation between half life and rate constant for a first order reaction.

(v) Draw neat and labelled diagram of dry cell.

(vi) Explain the structure of sulphur dioxide.

(vii) What is calcination? Explain it with reactions.

(viii) Arrange the following reducing agents in the order of increasing strength under standard state conditions. Justify the answer.

Element	Al(s)	Cu(s)	Cl^{-} (aq)	Ni(s)
E^{o}	−1.66 V	0.34 V	1.36 V	−0.26 V

Solution 2:

(i) (a) **Enthalpy of fusion:** The enthalpy change or amount of heat absorbed that accompanies the fusion of mole of a solid at constant temperature and pressure is called enthalpy of fusion. It is denoted by

$(\Delta_{fus}H)$.

(b) **Enthalpy of atomization:** The enthalpy change accompanying the dissociation of all the molecule in one mole of a gas phase substance into gaeous atom is called enthalpy of atomization. It is denoted by

$(\Delta_{ato}H)$.

(ii) Consider a dilute solution of a volume V (in dm^3) containing W grams of a substance having molecular weight M at an absolute temperature T. Then, concentration of the solution, C=n/V and No. of moles of the substance, n = W/M.

By van't Haff Boyle's law, at constant temperature, the osmotic pressure of a solution is directly proportional to its molar concentration or inversely proportional to the volume of the solution. Thus,

$\pi \alpha C$ (when temperature is constant)

Van't Hoff Charle's law at the constant concentration the osmotic pressure of a dilute solution is directly proportional to the absolute temperature (T), ie,

$\pi \alpha T$

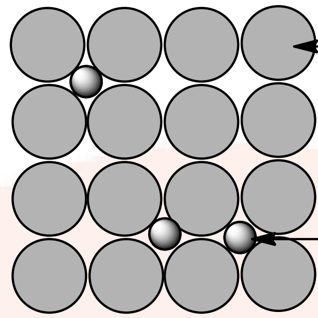
$\pi \alpha CT$

$\pi = CRT$

The equation is called vant's Hoff general solution equation.

R is constant of proportionality, also called general solution constant or gas constant.

(iii) The impurity defect found in stainless steel is interstitial impurity defect. In interstitial impurity defect, the impurity of cation is present in the interstitial position and make crystal defected. Stainless steel is an alloy of Iron and% Chromium mixed with it. Stainless steel typically contains about 1% carbon, 1-5% Manganese, 0.05% Phosphorous, 1-3% Silicon, 5% -10% Nickel and 15% - 20% Chromium. Carbon is a second – period element that is non-metallic and much smaller than iron. Carbon will therefore tend to occupy interstitial sites in the iron lattice.



Stainless steel

(iv) For a first order reaction,

$$t = \frac{2.303}{k}\log\frac{[R]_0}{[R]}$$

Where, $[R]_0$ is intial concentration and [R] is the final concentration.

At

$$t = t_{1/2} [R] = \frac{[R]_0}{2}$$

$$t_{1/2} = \frac{2.303}{k}\log\frac{[R]_0}{[R]_0/2}$$

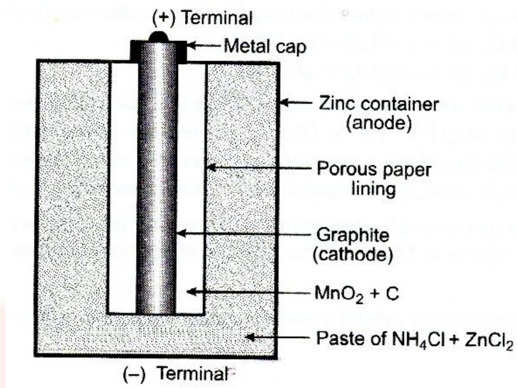
$$= \frac{2.303}{k}\log 2$$

$$t_{1/2} = \frac{2.303}{k}\times 0.3010$$

$$t_{1/2} = \frac{0.693}{k}$$

Therefore, $t_{1/2}$ for a first order reaction is independent of initial concentration.

(v) **A neat labelled diagram of Dry cell:**



(vi) **Structure of sulphur dioxide:**

(a) The structure of SO_2 molecule is V-shaped with S-O-S bond angle 119^o and bond dissociation enthalpy is 297 kJ/mol.

(b) In SO_2, each oxygen atom is bounded to sulphur by a σ and a π bond and the σ bonds between S and O are formed by sp^2-p overlapping.

(c) Sulphur in SO_2 is sp^2 hybridised forming three hybrid orbitals. Due to lone pair electrons, bond angle is reduced from 120^o to 119^o.

(d) One of π bonds is formed by $p\pi$- $p\pi$ overlapping while other π bond is formed by $p\pi$-$d\pi$ overlap.

(vii) **Calcination:** It is a process in which the ore is heated to a high temperature below the melting point of metal in absence of air or limited supply of air.

The changes that takes place during calcination with reactions are:

(a) Moisture and water from hydrated ores, volatile impurities and organic matter are removed.
(b) Ore becomes porous.
(c) Carbonates decompose to oxides.

$$ZnCO_3 \xrightarrow{\Delta} ZnO + CO_2$$

$$CuCO_3 \xrightarrow{\Delta} CuO + CO_2$$

$$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$$

$$\underset{\text{(Dolomite)}}{CaCO_3.MgCO_3} \xrightarrow{\Delta} CaO + MgO + 2CO_2$$

$$2Fe_2O_3.3H_2O \xrightarrow{\Delta} 2Fe_2O_{3(s)} + 3H_2O$$

(viii) The following arrangements shoes the order of increasing strength of the given reducing agents under standard state conditions;

$$\underset{-1.66V}{Al(s)} > \underset{-0.26V}{Ni(s)} > \underset{0.34V}{Cu(s)} > \underset{1.36V}{Cl(aq)}$$

Q.3. Answer any THREE of the following: [9]

(i) Determine whether the reactions with the following ΔH and ΔS values are spontaneous or non-spontaneous. State whether the reactions are exothermic or endothermic.

a. $\Delta H = -\ 110$ kJ, $\Delta S = +\ 40\ J\ K^{-1}$ at 400 K

b. $\Delta H = +\ 40$ kJ, $\Delta S = -\ 120\ J\ K^{-1}$ at 250 K

(ii) 1.0×10^{-3} kg of urea when dissolved in 0.0985 kg of a solvent, decreases freezing point of the solvent by 0.211 K. 1.6×10^{-3} kg of another non-electrolyte solute when dissolved in 0.086 kg of the same solvent depresses the freezing point by 0.34 K. Calculate the molar mass of the another solute.

(Given molar mass of urea = 60)

iii. Sucrose decomposes in acid solution into glucose and fructose according to the first order rate law with $t_{1/2}$ = 3 hours. What fraction of the sample of sucrose remains after 8 hours?

(iv) Explain how does nitrogen exhibit anomalous behaviour amongst group 15 elements.

Solution 3:

(i) (a) Given:

$\Delta H = -110kJ$

$\Delta S = 40Jk^{-1}$

$= 0.04KJk^{-1}$

$\text{Temperature}, T = 4000K$

$\Delta G = ?$

Since, ΔH is –Ve, the reaction is exothermic.

$\Delta G = \Delta H - T\Delta S$

$= -110 - 400 \times 0.04$

$= -110 - 16$

$= -126kJ$

Since, ΔG is negative, the reaction is spontaneous and exothermic.

(b) Given:

$\Delta H = 40kJ$

$\Delta S = -120Jk^{-1}$

$= 0.12KJk^{-1}$

$\text{Temperature}, T = 250K$

$\Delta G = ?$

Since, ΔH is positive, the reaction is endothermic.

$\Delta G = \Delta H - T\Delta S$

$= 40 - 250 \times (-0.12)$

$= 40 + 30$

$= 70kJ$

Since, ΔG is positive, the reaction is non- spontaneous and endothermic.

(ii) Given: Mass of solute, $\text{Urea} = W_2 = 1.0 \times 10^{-3} kg = 1g$

Mass of solvent $= W_2 = 0.0985kg = 98.5g$

$\therefore \Delta T = 0.211k$

$\text{Molar mass of urea}(NH_2CONH_2) = M_2 = 60g / mol$

Mass of unknown substance $= W_2^{'} = 1.6 \times 10^{-3} kg = 1.6g$

Mass of a Solvent $= W_1^{'} = 0.086kg$

$\therefore \Delta T_F = 0.34k$

Molar mass of unknown substance $= M_2^{'} = ?$

This problem will be solved in two parts:

(I) To calculate molal depression constant ‘k_f’.
For urea solution:

$$\Delta T_F = K_f \times \frac{W_2 \times 1000}{W_1 \times M_2}$$

$$K_f = \frac{\Delta T_F \times W_1 \times M_2}{W_2 \times 1000}$$

$$= \frac{0.211 \times 98.5 \times 60}{1 \times 1000}$$

$$= \frac{1247.01}{1000}$$

$$= 1.24701$$

$$\Rightarrow K_F = 1.2 g / mol$$

(II) To calculate molar mass, M_2
For unknown solution

$$M_2^{'} = K_f \times \frac{W_2^{'} \times 1000}{W_1^{'} \times M_2}$$

$$\Rightarrow K_f = \frac{1.2 \times 1.6 \times 1000}{86 \times 0.4}$$

$$= \frac{1920}{29.24}$$

$$\therefore M_2^{'} = 65.66 g / mol$$

Hence, the molar mass of another solution is 65.66g/mol.

(iii) Given:

$$t_{1/2} = 3 hours,$$

$$t = 8 hours$$

As sucrose decomposes according to first order rate law,

$$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]_t}$$

Using the formula,

$$k = \frac{0.693}{t_{1/2}}$$

$$\therefore k = \frac{0.693}{t_{1/2}} = \frac{0.693}{3} = 0.231 hr^{-1}$$

From formula,

$$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]_t}$$

$$\therefore 0.231 = \frac{2.303}{8} \log \frac{[A]_0}{[A]_t}$$

$$\Rightarrow \log \frac{[A]_0}{[A]_t} = 0.8024$$

Taking antilog on both sides,

$$\frac{[A]_0}{[A]_t} = \text{Anti}\log(0.8024) = 6.345$$

$$\Rightarrow \frac{[A]_0}{[A]_t} = \frac{1}{6.345} = 0.158$$

Hence, the fraction of the sample of sucrose that remains after 8 hours is 0.158.

(iv) Anomalous behavior of nitrogen among group 15 elements:

Nitrogen is the first amongst gropu 15 elements, which has,

(a) The smallest atomic size.
(b) Highest electronegativity.
(c) The highest ionisation energy.
(d) Absence of d- orbitals.

The difference in the nature between nitrogen and other elements of 15 group are:

(a) Nitrogen is a gas while all the other elements are solid.
(b) Nitrogen exists as diatomic molecule, (N_2) while other elements exists as tetra atomic molecules (P_4, As_4, Sr_4, etc.)
(c) Being the highest electronegative element in group 15 nitrogen forms hydrogen bonding while other doesn't.
(d) Nitrogen forms pπ- pπ multiple bonds while other elements in the group from pπ-dπ multiple bonds.
(e) NH_3 is basic while other hydrides are very less basic.
(f) The trihalides of nitrogen (except NF_3) are unstable while the trihalides of the other elements are comparatively stable.
(g) Except nitrogen, all other elements have vacant d- orbitals, and due to this nitrogen doesn't undergo formation of coordination compound.

Q.4. **Answer any ONE of the following:**
[7]

(i) Niobium crystallises as body centred cube (BCC) and has density of 8.55 kg dm^{-3}. Calculate the atomic radius of niobium. (Given: Atomic mass of niobium = 93)

Write one statement of first law of thermodynamics and its mathematical expression.

Write the reactions involved in the zone of reduction in blast furnace during extraction of iron.

(ii) Write molecular formulae and structures of the following compounds:

(a) Dithionic acid

(b) Peroxymonosulphuric acid

(c) Pyrosulphuric acid

(d) Dithionous acid

Calculate E_{cell} and ΔG for the following at 28 °C:

$Mg_{(s)} + Sn^{2+}$ (0.04 M) → Mg^{2+} (0.06 M) + $Sn_{(s)}$

$E^o_{cell} = 2.23$ V

Is the reaction spontaneous?

Solution 4:

(i) Density of Niobium (Nb) crystal is = 8.55 kg/dm^3.
Crystalline structure is bcc.
Atomic mass no. of Nb = 93 g/mol
Avogadro number = NA = 6.022 ×10^{23} mol^{-1}
Atomic radius of Niobium = ?
In bcc unit cells, there are 8 atoms at 8- corners and 1 atom at the body centre.

Thus, Number of Nb atoms = $\frac{1}{8}\times 8+1=1+1=2$

Mass of 1 Nb atom =

$\frac{a^2}{6.022\times10^{23}}$

$=1.544\times10^{-22}\,kg$

Thus, Mass of 2Nb atom

$=2\times1.544\times10^{-22}$

$=3.088\times10^{-22}\,kg$

Mass of unit cells = Mass of 2Nb atom = 3.088 × 10^{-22} g
If a is the edge length of bcc unit cell.
Volume of unit cell = a^3

$$\text{Density} = \frac{\text{Mass of unit cell}}{\text{Volume of unit cell}}$$
$$\Rightarrow d = \frac{3.088\times10^{-22}}{a^3}$$
$$\Rightarrow a^3 = \frac{3.088\times10^{-22}}{d}$$
$$= \frac{3.088\times10^{-22}}{8.55}$$
$$= 0.361\times10^{-22}\,dm^3$$
$$= 36.1\times10^{-24}\,dm^3$$
$$\Rightarrow a = \left(36.1\times10^{-24}\right)^{1/3}$$
$$= 3.33\times10-8dm$$

If r is the radius of 1Nb atom, then in bcc structure

$$r = \frac{\sqrt{3}}{4}a = \frac{\sqrt{3}}{4}\times3.33\times10^{-8}$$
$$= 1.43\times10^{-8}\,dm$$
$$\approx 14.3nm(\text{where}, 1dm = 10^{-9}\,m)$$

Hence, the atomic radius of niobium is 14.3 nm.

The first law of thermodynamics states that:

Energy can neither be created nor be destroyed. The total energy of an isolated system(cannot exchange energy with its surroundings) remains the same.

Mathematical expression:

In a system that exchange both heat and mechanical work with its surroundings, the total energy or the energy change is described by taking both heat – and work in it's account.

$$U_2 - U_1 = \Delta U = q + W$$
$$q = \Delta U - W$$
$$\Delta U = q + W$$

This is the mathematical expression for first law of thermodynamics, from the point of view of conservation of energy. This expression justifies first law of thermodynamics.

Where, U_2 and U_1 = Energies of the system in the final and initial states, respectively.

ΔU = Change in internal energy

q = heat absorbed or evolved by the system.

W = Work done by the system or on the system.

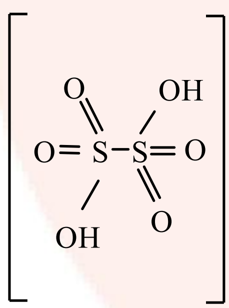
Reactions involved in the zone of reduction:

$$3Fe_2O_3 + CO \rightarrow 2Fe_3O_4 + CO_2$$
$$Fe_3O_4 + CO \rightarrow 3FeO_4 + CO_2$$
$$FeO + CO \rightarrow Fe + CO_2$$

(ii) (a) **Dithionic acid:**

Formula: $H_2S_2O_6$

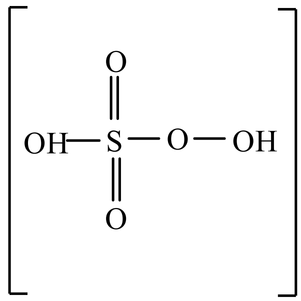
Structure:



(b) **Peroxy monosulphuric acid:**

Formula: H_2SO_3

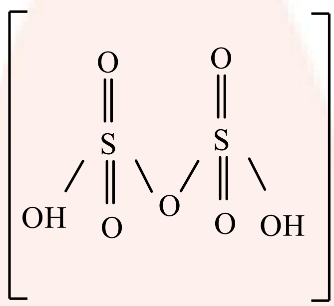
Structure:



(c) **Pyrosulphuric acid:**

Formula: $H_2S_2O_7$

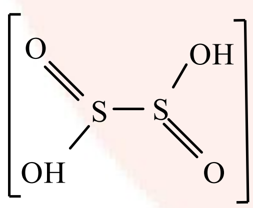
Structure:



(d) **Di-thionous acid:**

Formula: $H_2S_2O_4$

Structure:



Given: Cell reaction

$$Mg_{(s)} + Sn^{2+}{}_{(aq)} \rightarrow Mg^{2+}{}_{(aq)} + Sn_{(s)}$$

$E^{o}_{cell} = 2.23; [Sn^{2+}] = 0.04M$

$[Mg^{2+}] = 0.06M; n = 2$

$E_{cell} = E^{o}_{cell} - \frac{0.0592}{2}\log_{10}\frac{[Mg^{2+}]}{[Sn^{2+}]}$

$= 2.23 - \frac{0.0592}{2}\log_{10}\frac{[0.06]}{[0.04]}$

$= 2.23 - 0.296 \cdot \log_{10}[1.5]$

$= 2.23 - 0.296(0.1760)$

$= 2.23 - 0052096$

$\Delta G = nFE_{cell}$

$= -2 \times 96500 \times 2.224$

$= 429.232J$

$= -429.2KJ$

Since, ΔG for a given cell reaction is negative, the given reactants is spontaneous.

SECTION – II

Q. 5. **Select and write the most appropriate answer from the given alternatives for each sub-question:** **[7]**

(i) Identify the product 'D' in the following sequence of reactions:

$$H_3C - CH_2 - CH_2 - Cl \xrightarrow[KOH]{Alc} 'B' \xrightarrow{HBr} 'C' \xrightarrow[Ether]{Na} 'D'$$

(a) 2,2-dimethylbutane (b) 2 ,3-dimethylbutane (c) Hexane (d) 2 ,4-dimethylpentane

(ii) Which of the following complexes will give a white precipitate on treatment with a solution of barium nitrate?

(a) $[Cr(NH_3)_4SO_4]Cl$ (b) $[Co(NH_3)_4Cl_2]NO_2$

(c) $[Cr(NH_3)_4Cl_2]SO_4$ (d) $[CrCl_2(H_2O)_4]$ Cl

(iii) What is the geometry of chromate ion?

(a) Tetrahedral (b) Octahedral (c) Trihonal planer (d) linear

(iv) Primary and secondary nitroalkanes containing α-H atom show property of ________.

(a) Chain isomerism (b) tautomerism (c) Optical isomerism (d) geometrical isomerism

(v) In phenol carbon atom attached to −OH group undergoes ________.

(a) sp^3 hybridisation (b) sp hybridization (c) sp^2 hybridisation (d) no hybridization

(vi) Identify the strongest acid amongst the following.

(a) Chloracetic acid (b) Acetic acid (c) Trichloroacetic acid (d) Dichloroacetic acid

(vii) Which of the following vitamins is water soluble?

(a) A (b) D (c) E (d) B

Solution 5:

(i) (c) Hexane

(ii) (c) $[Cr(NH_3)_4Cl_2]SO_4$

(iii) (a) Tetrahedral

(iv) (b) Tautomerism

(v) (c) sp^2 hybridisation

(vi) (c) Trichloroacetic acid

(vii) (d) B

Q.6. Answer any SIX of the following: [12]

(i) Write a note on Friedel Craft's acylation.

(ii) How is ethylamine prepared from methyl iodide?

(iii) What are antibiotics? Give 'two' examples.

(iv) Explain, why are boiling points of carboxylic acids higher than corresponding alcohols.

(v) How are proteins classified on the basis of molecular shapes?

(vi) What are interstitial compounds? Why do these compounds have higher melting points than corresponding pure metals?

(vii) Write the structures and IUPAC names of the following compounds:

(a) Adipic acid

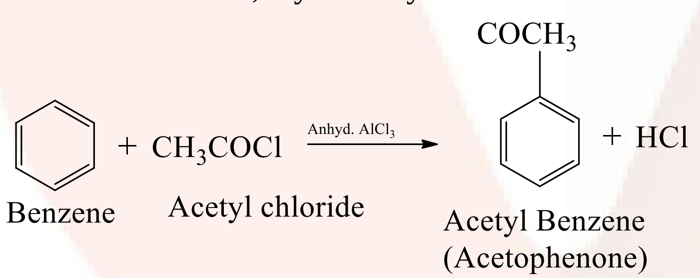
(b) α- methyl butyraldehyde

(viii) Explain with examples, branched and linear polymers.

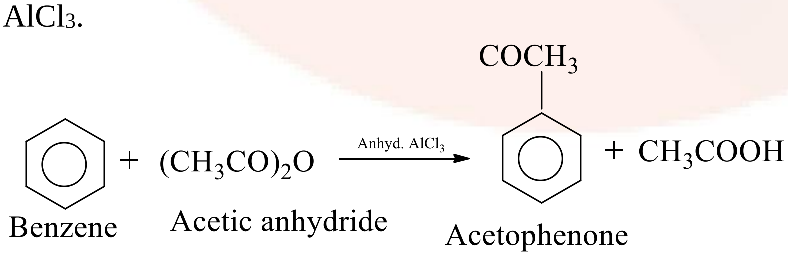
Solution 6:

(i) **Friedel Craft's acylation:** Friedel Craft's acylation attaches an acyl group (-COR) to a benzene ring.

For example: When Benzene is treated with acyl halide in the presence of anhydrous aluminium chloride, it yields acylbenzene.



This reaction is called **Friedel Craft's reaction.**
Acylation can also be carried out with acetic anhydride in the presence of anhydrous $AlCl_3$.



(ii) The preparation of ethyl amine from methyl iodine is done in two steps:
Step 1: Methyl iodine when react with potassium cyanide forms Acetonitrile.

$$\underset{\text{Methyl Iodine}}{CH_3-I} + \underset{\text{Potassium Cynaide}}{KCN} \rightarrow \underset{\text{Acetonitrile}}{CH_3CN} + \underset{\text{Potassium Iodine}}{KI}$$

Step 2: In the second step, acetonitrile further reacts with hydrogen to form ethyl amine in presence of (Na) sodium and ethyl alcohol.

$$\underset{\text{Acetonitrile}}{CH_3CN} \xrightarrow{Na/C_2H_5OH} \underset{\text{Ethyl amine}}{CH_3CH_2NH_2}$$

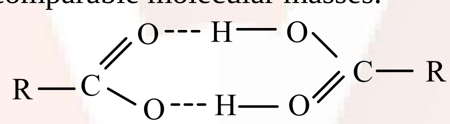
(iii) **Antibiotics:** Antibiotics is a drug derived from living matter or micro- organism, used to kill or prevent the growth of other micro- organism.

(i) Board spectrum antibiotics: Chloramphenicol, Ampicillin.

(ii) Narrow spectrum antibiotics: Penicillin.

(iv) The boiling point of carboxylic acids higher than corresponding alcohols because of their ability to form intermolecular hydrogen bonding. The hydrogen bond formed by the carboxylic acids are stronger than those in alcohols because O-H bond in COOH is more strongly polarized sue to the presence of electron withdrawing carboxy group in adjacent position then the O-H bond

of alcohols. Therefore, the boiling points of carboxylic acids particular lower members, are higher than alcohol of comparable molecular masses.



Intermolecular hydrogen bonding in carboxylic acid.

(v) Classification of proteins on the basis of molecular shapes are:

Fibrous proteins: The proteins in which the polypeptide chains lie parallel (side by side) to from fibre- like structure are called fibrous proteins. The polypeptide chains held together by hydrogen bonds. These proteins are insoluble in water.

The fibrous proteins are tough and insoluble in water, dilute acids or bases. Example: myocin (in muscles), keratin(in a hair, nails, skin), fibroin (in silk), collagen(in tendons), etc.

Globular proteins: The proteins are folded to form spherical structure and have intramolecular hydrogen bonding are called globular proteins. They are soluble in water and dilute acids or bases.

Example: Hemoglobin (in blood), albumin (in eggs), insulin (in pancreas), etc.

(vi) **Interstitial compounds:** They are those compounds of the transition metals which are formed when small atoms like H, C or N are trapped inside the interstitial vacant spaces in the crystal lattices of the metals. Some of the examples of interstitial compounds are TiC, $TiH_{1.73}$, Mn_4N, Fe_3H, etc.

The interstitial compounds have high melting point due to the following reasons:

(i) They are very hard.

(ii) They are chemically insert.

(iii) The presence of the foreign atoms stabilizes the crystal structure.

(vii) Structure and IUPAC name of the compounds are:

(a) **Adipic acid**:

Structure:

$$OH-\underset{\underset{O}{\|}}{C}-CH_2-CH_2-CH_2-CH_2-\overset{\overset{O}{\|}}{C}-OH$$

IUPAC Name: Hexanedioic acid.

(b) α **-methyl butyraldehyde:**

Structure:

$$CH_3-CH_2-\underset{\underset{CH_3}{|}}{CH}-COH$$

IUPAC name: 2-methyl butyraldehyde.

(viii) **Linear polymers:** The polymers made up of long continuous chain without any branches are called linear polymers. They have high melting points, high densities and high tensile strength. Example: Polyethene, PVC.

Branched chain polymers: The polymers made up of main chain (linear chain) with smaller chains as branches of main chain are called branched chain polymers. They have lower melting points, densities and tensile strength as compared to linear polymers. Example: Polypropylene.

Q 7. Answer any THREE of the following:
[9]

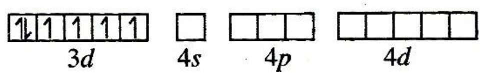
(i) On the basis of valence bond theory explain the nature of bonding in $[CoF_6]^{3-}$ ion. Write the IUPAC name of $[Co(NO_2)_3(NH_3)_3]$.

(ii) Define lanthanoid contraction. Explain its effects.

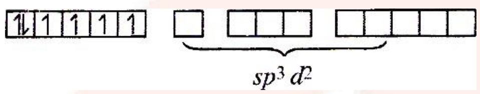
(iii) Write mechanism of Aldol addition reaction.

(iv) Define carbohydrates. What are reducing and non-reducing sugars?

Solution 7:
(i) $[CoF_6]^{3-}$ Cobalt exists in the +3 oxidation state.
Orbitals of Co^{3+} ion:



Again, fluoride ion is a weak field ligand. It cannot cause the pairing of the 3d electrons. As a result, the Co^{3+} ion will undergo sp^3d^2 hybrization, sp^3d^2 hybridized orbitals of Co^{3+} ions are:



Hence, the geometry of the complex is octahedral and paramagnetic.

(ii) Lanthanoid contraction may be defined as gradual decrase in atomic and ionic radii of lanthanoids with the increase in atomic number.

Effects of lanthanoid:

Decrease in basicity:
(i) In lanthaoids due to Lanthanoid contraction, as the atomic number increases, the size of the lanthanoid atoms and their tripositive ions decreases, ie, from La^{+} to Lu^{3+}.
(ii) As size of the cation decreases, according to Fajan's rule, the polarizability character decreases. Therefore, the basic nature of the hydroxides decreases.
(iii) Basicity and ionic character decrases in the order $La(OH)_3 > Ce(OH)_3 >$…..$Lu(OH)_3$.

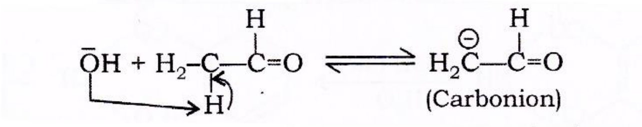
Ionic Radii of post- lanthanoid

(i) Due to lanthanoid contraction the atomic radii(size) of elemnts which follow lanthanum in the 6th period (3rd transition series- Hf, Ta, W, Re)- are similar to the elements of the 5th period (4d- series zr, Nb, Mo, Tc).
(ii) Due to similarity in their size, post- lanthanoid elements (5d-series) have closely similar properties to the elements of the 2nd transition series (4d- series) which lie immediately above them.

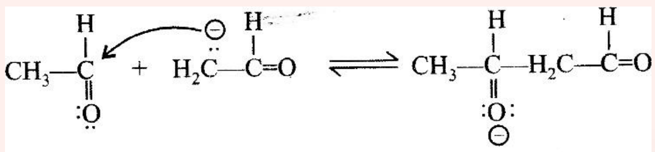
(iii) Pairs of elements namely Zr-Hf(Gr-4), Nb-Ta(GR-5), Mo-W(Gr-6), Tc-Re(Gr-7) are called chemical twins since they possess almost identical sizes and similar properties.

(iii) **Mechanism of Aldol addition reaction:**

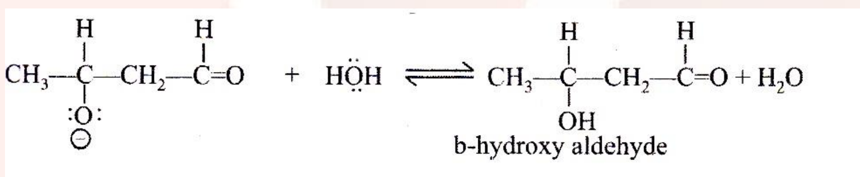
Step 1- Base (-OH-) abstracts a hydrogen atom from α-carbon of aldehyde to form carnanion. In this step hydroxide ion function as a base and removes the acidic α-hydrogen giving the reactive enolate.



Step 2- The nucleophillic enolate attacks the aldehyde at the electrophilic carbonyl C in a nucleophilic addition type process giving an intermediate alkoxide.



Step 3- Alkoxide ion abstracts a hydrogen ion from water to from β-hydroxy aldehyde. Base –OH^- ion is regenerated.



(iv) **Carbohydrates:** Carbohydrates are optically active polyhydroxy aldehydes or polyhydroxy ketons, or the compounds which on hydrolysis produce polyhydroxy aldehydes or polyhydroxy ketones.

Example: Glucose, Sucrose, fructose.

Reducing sugars:

Carbohydrates which reduces Fehling’s solution to red ppt. of Cu2O or Tollen’s reagent to shining metallic silver are called reducing sugars. All monosaccharides and oligosaccharides except sucrose are reducing sugars.

Non- reducing agent:

Carbohydrates which do not reduce Fehling’s solution and Tollen’s reagent are called non- reducing sugars, example, sucrose.

Q8. Answer any ONE of the following: [7]

(i) Write a note on Gabriel phthalimide synthesis.

What are biodegradable polymers and non-biodegradable polymers? Write ‘one example’ of each. Explain cationic detergents.

(ii) How is carbolic acid prepared from the following compounds?

(a) Aniline

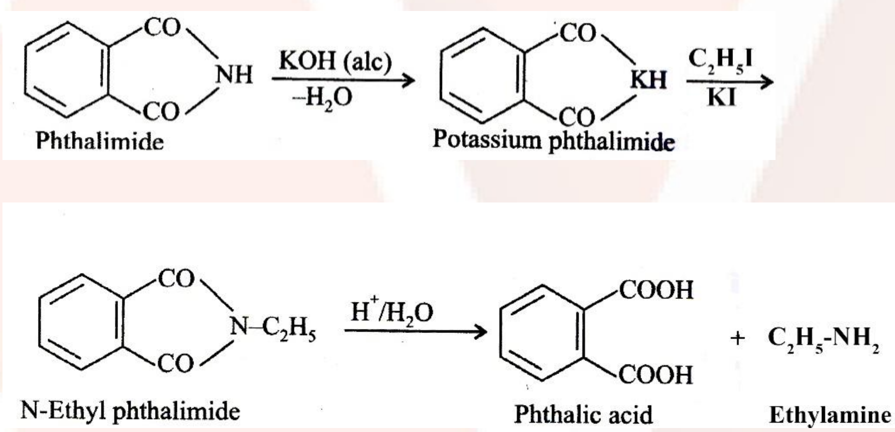
(b) Chlorobenzene and steam at 698 K?

Draw structure of DDT. Write its environmental effects. Mention 'two' physical properties of carbolic acid.

Solution 8:

(i) **Gabriel phthalimide synthesis:** This synthesis is useful to get pure amino acids in battery yield. Their reaction involves the preparation of primary amines free from secondary and tertiary amines by using the reaction between alkyl halides with alkali metal phthalimide and subsequent hydrolysis.

For example, phthalimide is firstly treated with alocohiloc KOH to from potassium phthalimide which on further reaction with ethyl iodide gives N-ethyl phthalimide =. The product N- ethylphthalimide is hydrolysed with dilute HCl to from primary amine, ie, ethyl amine. This reaction is known Gabriel phthalimide synthesis.



Biodegradable polymers: Polymers which disintegrate by themselves after a certain period of time due to environmental degradation are called biodegradable polymers.

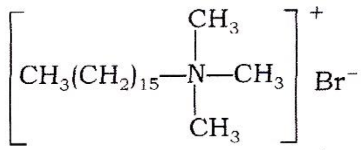
Non- biodegradable polymers: Polymers which do not disintegrate by themselves after a certain period are call non- biodegradable polymers.

Example of Biodegradable polymers: PHBV ie., Polyhydroxy butyrate- Co- β -hydorxy valerate Dextron.

Example of Non- Biodegradable polymers: Bakelite, Terylene, Nylon.

Cationic detergents:

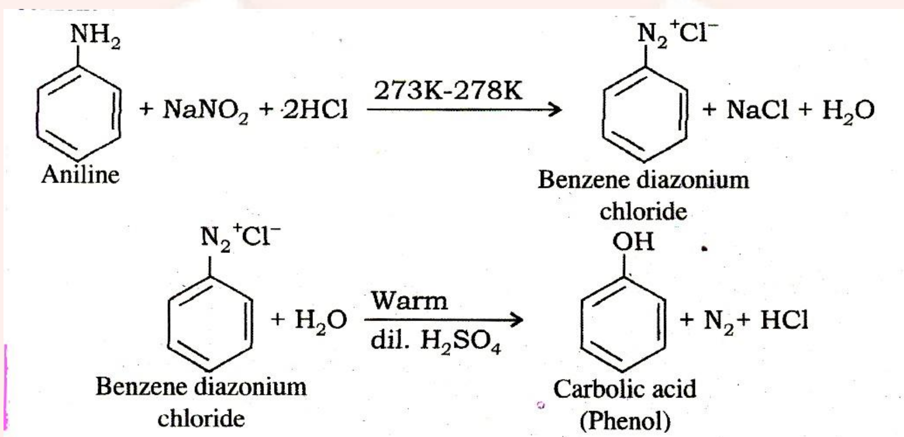
Cationic detergents are quaternary ammonium salts of amines with chloride, accetates or bromides as anions. Cations are long chain hydrocarbons and a positive charge on nitrogen atom. They are used as germicides, cetyltrimethyl ammonium chloride is used in hair conditioners.



***n*-hexadecyl terimethyl ammonium chloride**

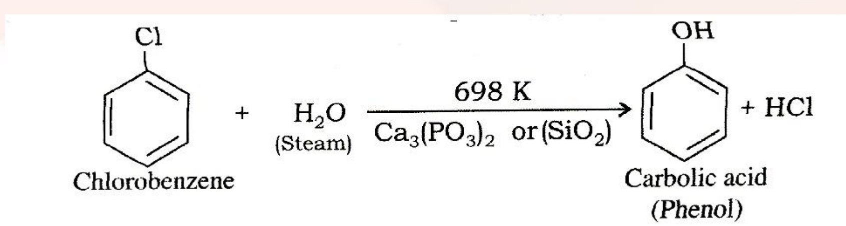
(ii) (i) **Preparation of carbolic acid from aniline:**

When aniline is treated with sodium nitrite and hydrochloric acid at low temperature (0°C-5°C), benzene diazonium chloride is formed. This reaction is called diazotization. An acquous solution of benzene diazonium chloride on warning with dil. H_2SO_4 gives carbolic acid or phenol

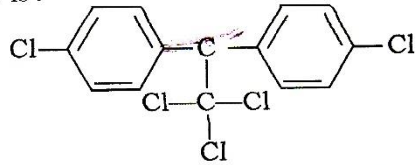


(ii) **Preparation of carbolic acid from chlorobenzene and steam at 698K:**

Cholrobenzene is treated with steam in the presence of calcium phosphate or silica(SiO_2) as a catalyst when carbolic acid or phenol is obtained.



The structure of DDT is :



Dichloro-diphenyl trichloro ethane (DDT)

Some of the environmental effects of DDT are:

1. It is not readily metabolized by animals.
2. It is deposited and stored in fatty tissues.

The physical properties of carbonic acid are:

1. Pure phenol is a colorless, hygroscopic, crystalline solid with typical phenolic color. On exposure to air and light, it turns pink due to partial oxidation.
2. Phenol has low melting point (316K), but has higher boiling point (455K) due to intermolecular hydrogen bonding.